

ONNEILE TLOUANA. 20230076. NET731. WEEK6. SECURITY REVIEW

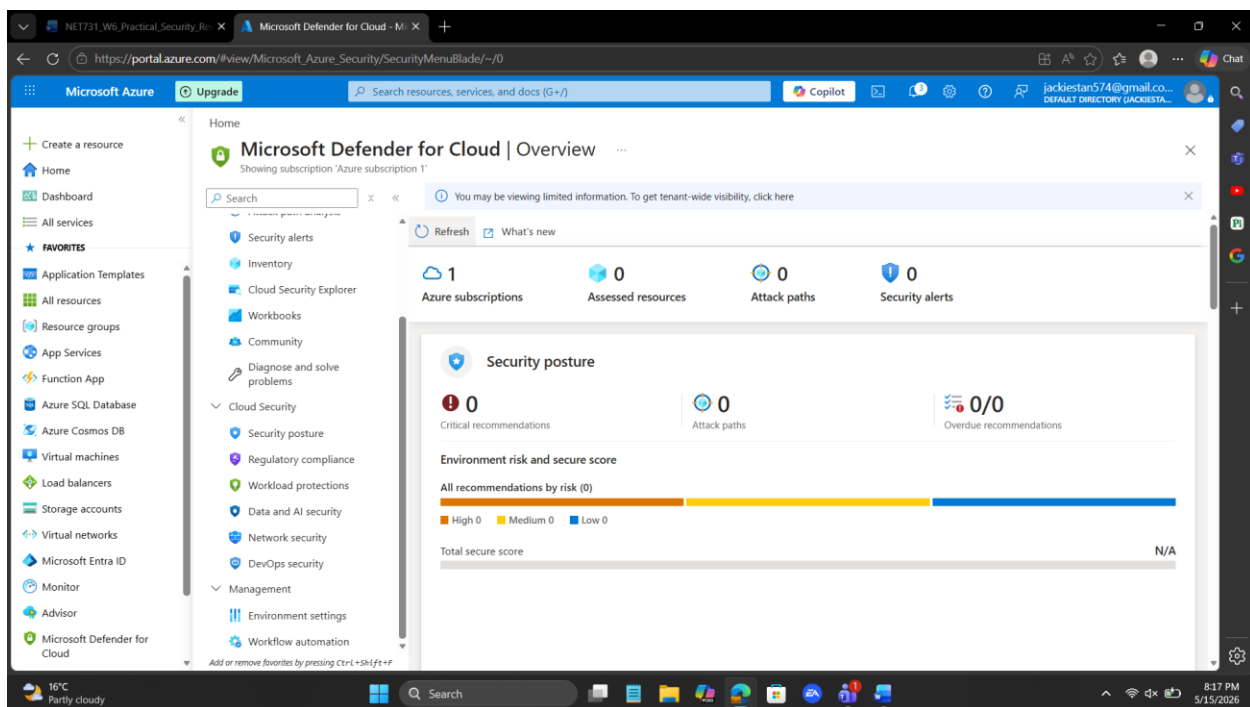
This practical focused on reviewing and improving the security posture of the Azure environment using Microsoft Defender for Cloud.

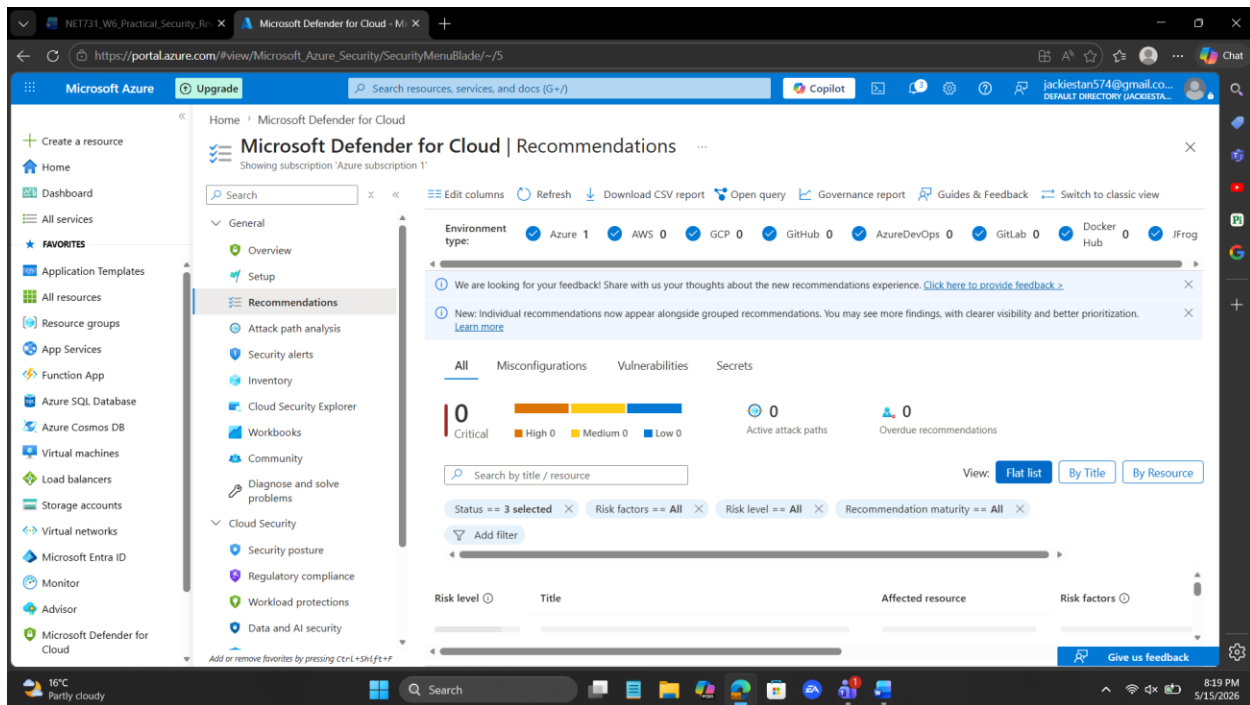
Security recommendations were reviewed and remediation actions were applied to reduce vulnerabilities and improve resource protection.

Below are screenshots of:

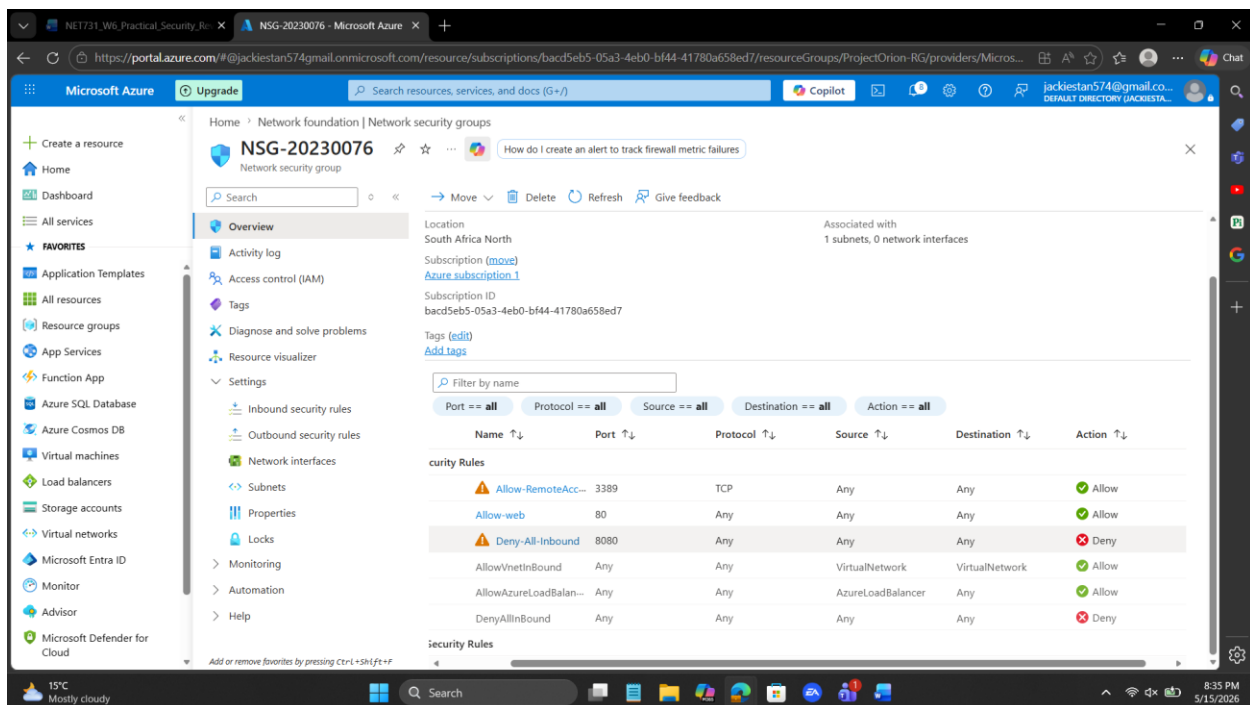
- Open Microsoft Defender for Cloud (Free Tier)
- Review of security recommendations
- Apply fixes (close open ports, enable OS updates)

Task 1, screenshots of Microsoft Defender





Task 2, Below are screenshots of closing ports in NSG



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NSG-20230076

Network security group

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Move Delete Refresh Give feedback

Overview

Location: South Africa North
Associated with: 1 subnets, 0 network interfaces

Subscription (move): Azure subscription 1
Subscription ID: bacd5eb5-05a3-4eb0-bf44-41780a658ed7
Tags (edit): Add tags

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Name	Port	Protocol	Source	Destination	Action
Allow-RemoteAccess	3389	TCP	102.253.65.11	Any	Allow
Allow-web	80	Any	Any	Any	Allow
Deny-All-Inbound	8080	Any	Any	Any	Deny
AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
AllowAzureLoadBalanc...	Any	Any	AzureLoadBalancer	Any	Allow
DenyAllInBound	Any	Any	Any	Any	Deny

Security Rules

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Allow-web	80	Any
Deny-All-Inbound	8080	Any
AllowVnetInBound	Any	Any
AllowAzureLoadBalanc...	Any	Any
DenyAllInBound	Any	Any

Allow-RemoteAccess

NSG-20230076

Source: Any

Source port ranges: *

Destination: Any

Service: RDP

Destination port ranges: 3389

Protocol: ☒ TCP

Action: Give feedback

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Microsoft Azure portal interface showing the configuration of a Network Security Group (NSG) rule named 'Allow-RemoteAccess'.

Overview:

- Location: South Africa North
- Subscription: [Subscription 1](#)
- Subscription ID: bacd5eb5-05a3-4eb0-bf44-41780a658ed7
- Tags: [Add tags](#)

Security Rules:

Name	Port	Protocol
Allow-RemoteAccess	3389	TCP
Allow-web	80	Any
Deny-All-Inbound	8080	Any
AllowVnetInbound	Any	Any
AllowAzureLoadBalanc	Any	Any
DenyAllInBound	Any	Any

Configuration Details:

- Source: My IP address
- Source IP addresses/CIDR ranges: 102.253.65.11
- Source port ranges: *
- Destination: Any
- Service: RDP
- Destination port ranges: 3389
- Protocol: TCP

Update Rule: Updating security rule 'Allow-RemoteAccess'.

Microsoft Azure portal interface showing the configuration of a Network Security Group (NSG) rule named 'Deny-All-Inbound'.

Overview:

- Location: South Africa North
- Subscription: [Subscription 1](#)
- Subscription ID: bacd5eb5-05a3-4eb0-bf44-41780a658ed7
- Tags: [Add tags](#)

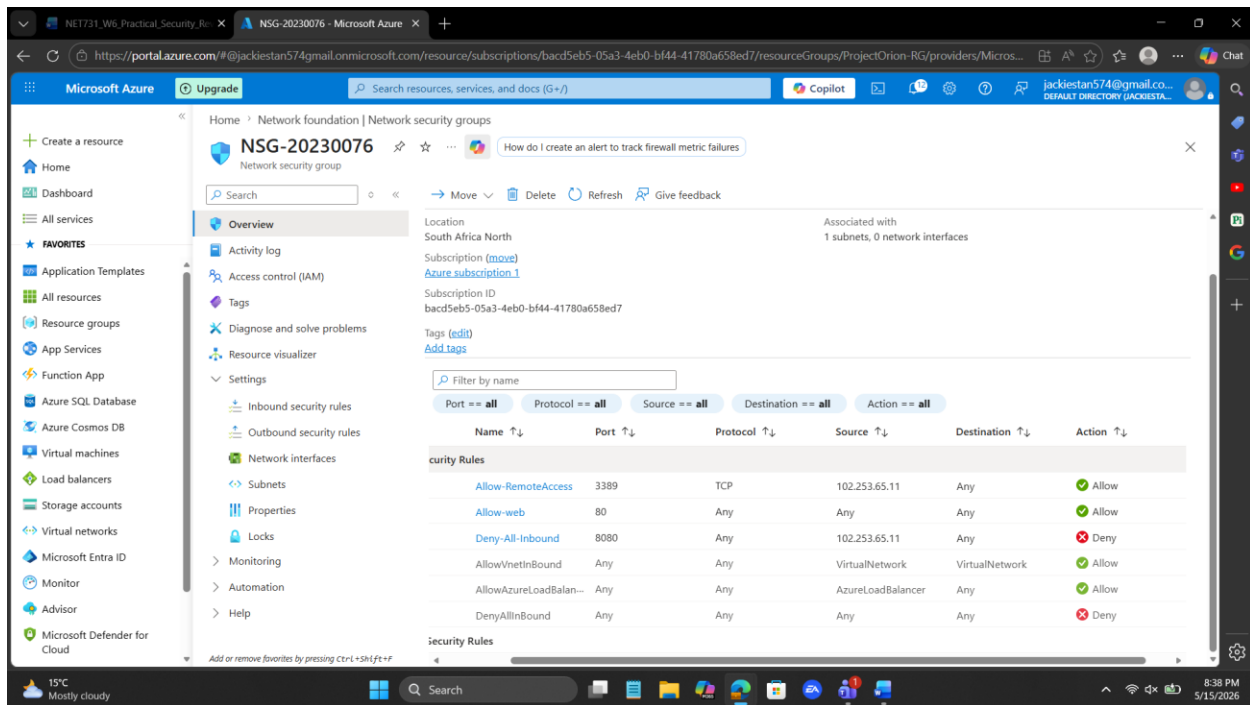
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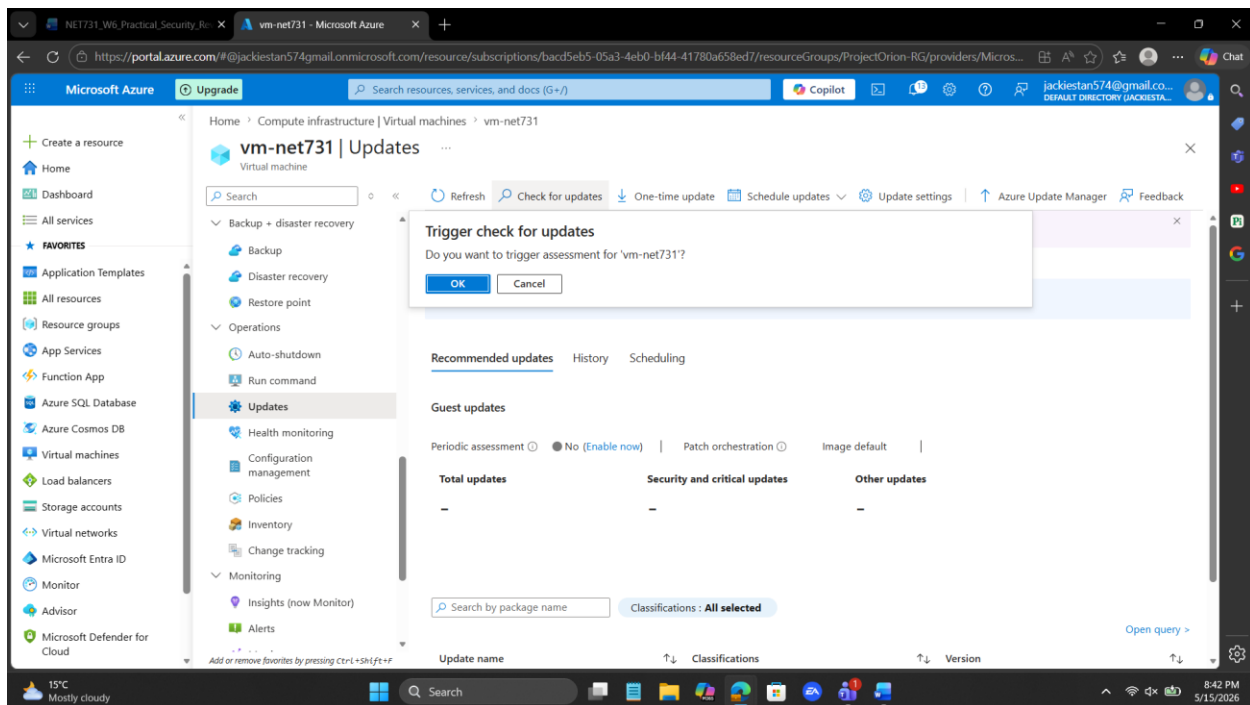
Configuration Details:

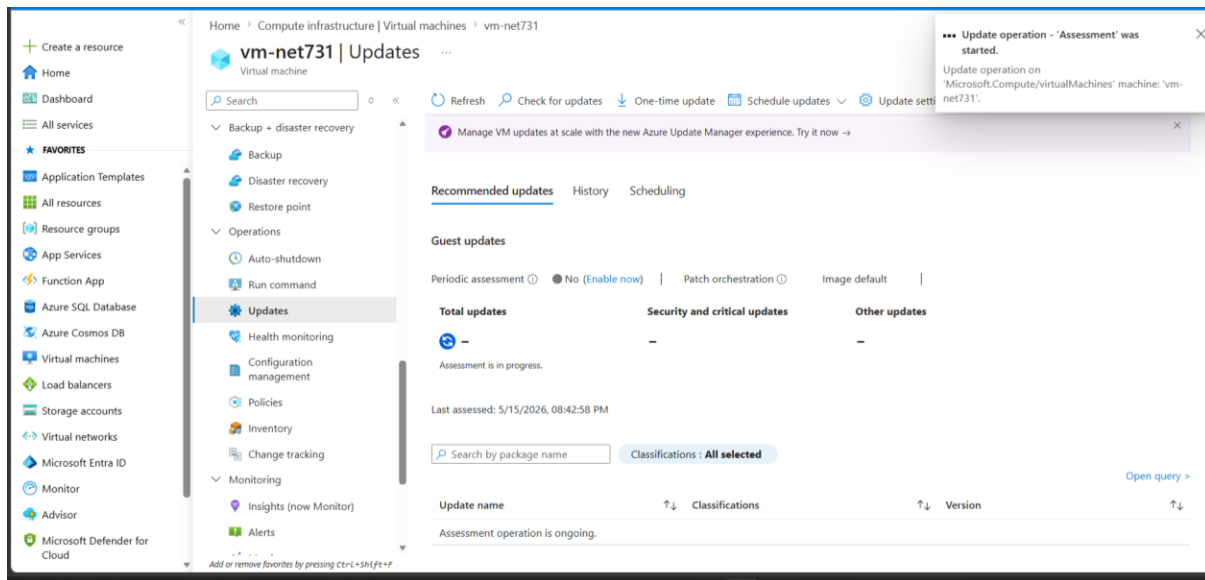
- Source: My IP address
- Source IP addresses/CIDR ranges: 102.253.65.11
- Source port ranges: *
- Destination: Any
- Service: Custom
- Destination port ranges: 8080
- Protocol: Any

Update Rule: Updating security rule 'Deny-All-Inbound'.



Task 3,Below are screenshots of enabling OS updates





SUMMARY

Part 1. Security findings

The environment had open remote management ports (RDP/SSH) exposed to the internet.

Part 2. Fix applied

Restricted NSG inbound rules to allow only trusted IP addresses.

Part 3. Update remediation

Enabled OS updates to patch vulnerabilities.

REFLECTION

This practical improved my understanding of cloud security auditing, vulnerability management and security remediation.

SKILLS DEMONSTRATED

- Microsoft Defender for Cloud
- Security Auditing
- Vulnerability Management
- Security Recommendations
- Cloud Security

